

Krish Mittal

Full-Stack & Product-Focused Developer | Software Engineering Student

📍 Karnal, Haryana, India 📞 +91-8307142813 ✉️ Krish4789.be23@chitkara.edu.in
🔗 <https://www.linkedin.com/in/krishh-mittal/>

Professional Summary

Product-minded Software Engineering student with strong foundations in full-stack web development, backend APIs, and data structures. Experienced in building scalable, user-centric applications and IoT-integrated systems using JavaScript, React, Node.js, Java (Spring Boot), and cloud-ready architectures. Adept at writing clean, maintainable code, optimizing performance, and owning features end-to-end in fast-paced, collaborative environments. Passionate about building high-impact software that solves real-world business problems.

Education

Bachelor of Engineering — Software Engineering

Chitkara University, Patiala, India

Expected Graduation: June 2027

Current Semester: 6

Relevant Coursework:

Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Web Development, Software Engineering (SDLC), System Design, Cloud Computing

Technical Skills

Programming Languages: JavaScript, Java, C#, C++

Web & Backend Technologies: React.js, Node.js, Express.js, ASP.NET Core, Spring Boot

Databases: MySQL, PostgreSQL, MongoDB, Firestore

Tools & Platforms: Git, Linux, macOS, Windows, REST APIs, WebSockets

Concepts: System Design, API Design, Authentication & Authorization, Real-Time Systems, IoT Integration, Clean Code Practices, Dsa

Projects

Library Management System : Full-Stack Application | Java Spring Boot & Node.js + React

- Designed and developed a complete **Library Management System** to manage books, users, and issue/return workflows.

- Built backend APIs using **Spring Boot (Java)** with role-based access control for Admin, Librarian, and Student roles.
 - Implemented a **Node.js + Express** backend alternative with RESTful services and JWT-based authentication.
 - Developed a responsive frontend using **React.js**, enabling real-time book availability, issue tracking, and user dashboards.
 - Integrated relational databases (MySQL/PostgreSQL) for transactional consistency and optimized queries.
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Food Spoilage Detection System : IoT Project | Raspberry Pi & Arduino Nano 33 IoT

- Built an **IoT-based food management system** to predict food spoilage using environmental sensor data.
 - Collected temperature, humidity, and gas sensor readings via **Arduino Nano 33 IoT**, processed using **Raspberry Pi**.
 - Implemented logic to analyze spoilage thresholds and notify users before food becomes unsafe.
 - Demonstrated real-world application of embedded systems, sensor integration, and data-driven decision making.
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Smart Gate Pass Management System: IoT + Full-Stack Application

- Developed a **hostel gate pass system** integrating IoT hardware with a web-based frontend.
 - Enabled students to apply for gate passes online; wardens approve/reject requests via an admin dashboard.
 - Integrated **RFID-based access control** at the gate to allow exit only after approval.
 - Used **Raspberry Pi and Arduino Nano 33 IoT** for hardware integration and secure validation.
 - Improved security, reduced manual effort, and digitized the complete leave approval workflow.
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Real-Time Task Management System: Backend-Focused Project | Node.js, WebSockets, MongoDB

- Built a real-time **task assignment and collaboration system** using **Node.js** and **WebSockets**.
 - Enabled users to assign tasks, update statuses, and receive live updates without page refresh.
 - Designed MongoDB schemas and implemented **database sharding** for scalability.
 - Developed backend APIs with authentication, authorization, and data validation.
 - Used basic HTML/CSS frontend to demonstrate backend-first, system-level design.
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Employability Skills & Leadership

- **Teamwork:** Collaborated effectively on group projects and academic coursework.
- **Problem Solving:** Applied analytical thinking to solve complex coding and algorithmic challenges.
- **Communication:** Strong written and verbal communication through presentations and technical documentation.
- **Engineering Discipline:** Clean code practices, testing mindset, structured and maintainable development.